

Module 3

EMTP-type

Nodal-Analysis-Based Modeling and Simulation

Most Important Topics and Concepts

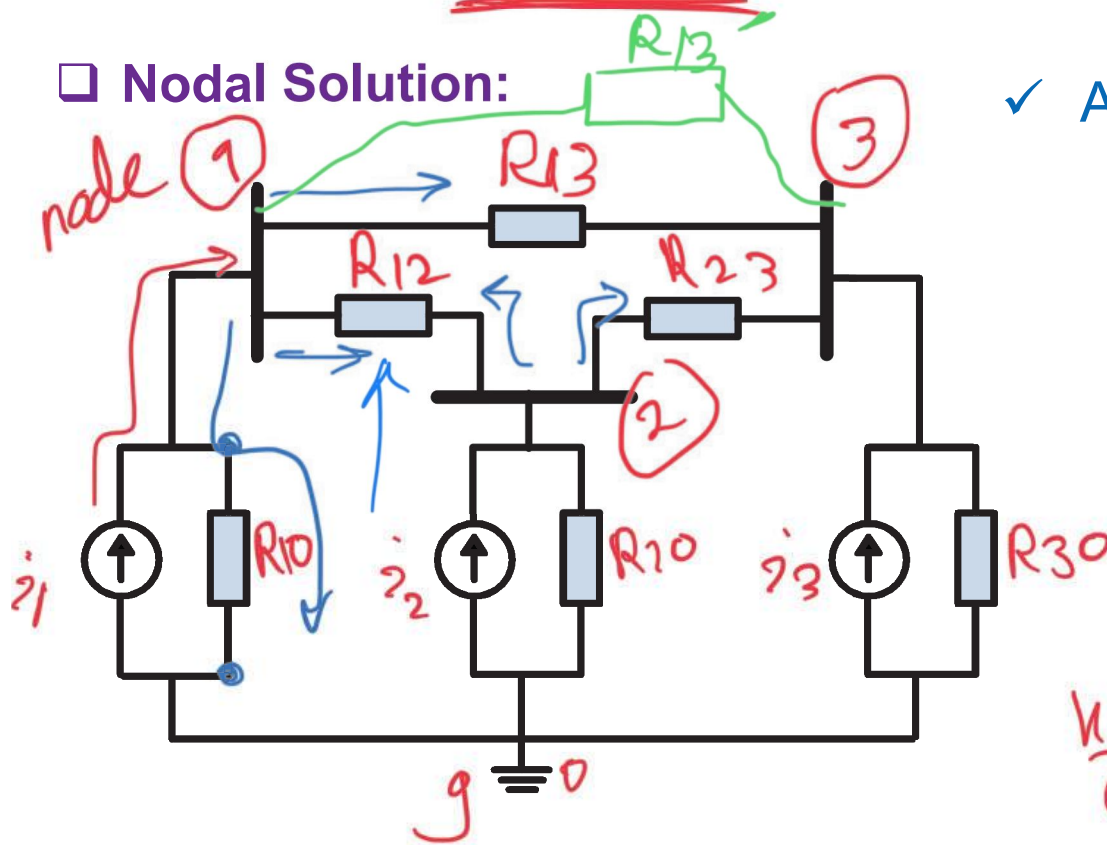
□ Principles of:

- Nodal EMT modeling of electric circuits
- Numerical stability and accuracy of integration methods
- Wave propagation and transmission line modeling
- Modeling of nonlinear components
- Nodal equation solution considerations

Nodal Circuit Modeling

□ Nodal Solution:

✓ Assume resistive branches



$$\text{KCL (1)}: i_1 = \frac{V_1}{R_{10}} + \frac{V_1 - V_3}{R_{13}} + \frac{V_1 - V_2}{R_{12}}$$

$$\text{KCL (2)}: i_2 = \frac{V_2}{R_{20}} + \frac{V_2 - V_1}{R_{12}} + \frac{V_2 - V_3}{R_{23}}$$

$$\text{KCL (3)}: i_3 = \frac{V_3}{R_{30}} + \frac{V_3 - V_1}{R_{13}} + \frac{V_3 - V_2}{R_{23}}$$

$$\begin{cases} i_1 = \left(\frac{1}{R_{10}} + \frac{1}{R_{13}} + \frac{1}{R_{12}} \right) V_1 + \left(-\frac{1}{R_{12}} \right) V_2 + \left(-\frac{1}{R_{13}} \right) V_3 \\ i_2 = \left(-\frac{1}{R_{12}} \right) V_1 + \left(\frac{1}{R_{20}} + \frac{1}{R_{12}} + \frac{1}{R_{23}} \right) V_2 + \left(-\frac{1}{R_{23}} \right) V_3 \\ i_3 = \left(-\frac{1}{R_{13}} \right) V_1 + \left(-\frac{1}{R_{23}} \right) V_2 + \left(\frac{1}{R_{30}} + \frac{1}{R_{13}} + \frac{1}{R_{23}} \right) V_3 \end{cases}$$

➤ Would the solution be true during dynamic conditions? yes!

$$\frac{1}{R_{ij}} = g_{ij}$$

Resistance

Conductance

$$g'_{13} = 1/R'_{13}$$

$$\begin{cases} (g_{10} + g_{13} + g_{12})v_1 + (-g_{12})v_2 + (-g_{13})v_3 = i_1 \\ (-g_{12})v_1 + (g_{20} + g_{12} + g_{23})v_2 + (-g_{23})v_3 = i_2 \\ (-g_{13})v_1 + (-g_{23})v_2 + (g_{30} + g_{13} + g_{23})v_3 = i_3 \end{cases}$$

$$\mathbf{G} \cdot \mathbf{V} = \mathbf{i}$$

$$\begin{bmatrix} g_{11} & -g_{12} & -g_{13} \\ -g_{12} & g_{22} & -g_{23} \\ -g_{13} & -g_{23} & g_{33} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix}$$

diagonal

known

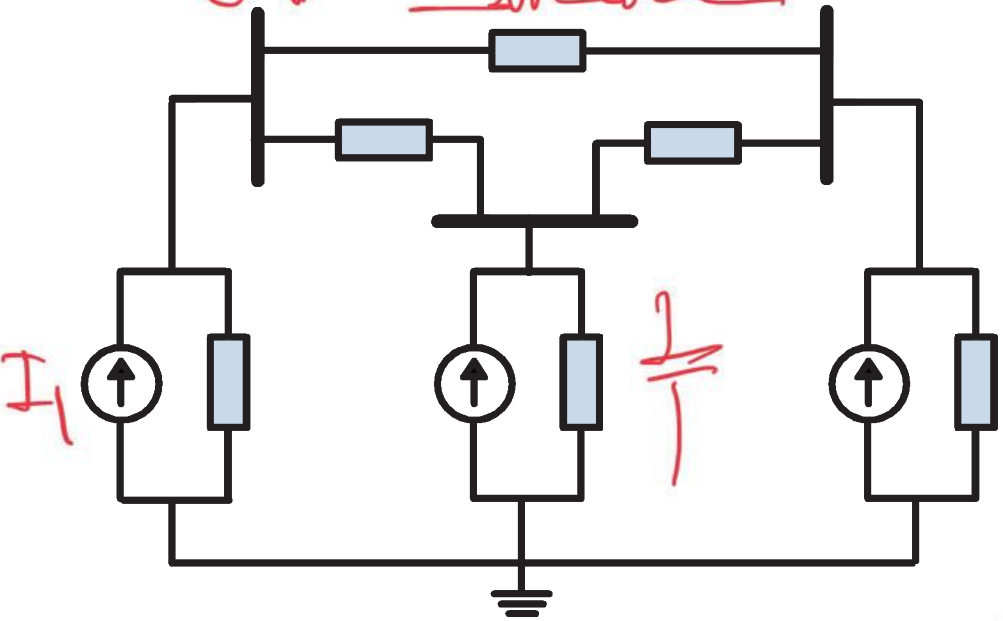
unknown

known

Nodal Circuit Modeling

□ Nodal Solution:

Assume impedances in branches



$V_i = |V_i| \angle \delta_i$
 $Y_{bus} \cdot V = I$
Assume Steady state!

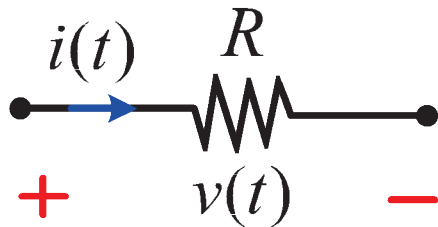
$Y_{11} = Y_{10} + Y_{12} + Y_{13}$
 $Y_{22} = Y_{20} + Y_{21} + Y_{23}$
 $Y_{33} = Y_{30} + Y_{31} + Y_{32}$

$$\begin{bmatrix} Y_{11} & -Y_{12} & -Y_{13} \\ -Y_{12} & Y_{22} & -Y_{23} \\ -Y_{13} & -Y_{23} & Y_{33} \end{bmatrix}$$

➤ Would the solution be true during dynamic conditions? **NO!**

Nodal Circuit Modeling

□ Discretization of Circuit Components:

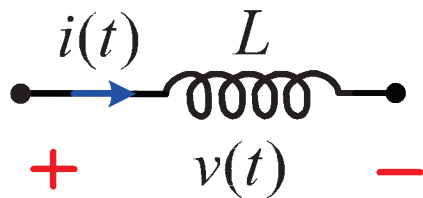


$$R = 2\Omega$$

$$i(10ms) = 10A$$

$$\Rightarrow V = RI$$

$$v(10ms) = 2 \times 10A = 20V$$



$$L = 10mH$$

$$i(10ms) = 10A$$



$$v(10ms) = ?$$

$$V(t) = L \frac{di(t)}{dt}$$

❖ Discretization of L:

$$\int_{t-\Delta t}^t L di(t) = \int_{t-\Delta t}^t V(t) dt$$

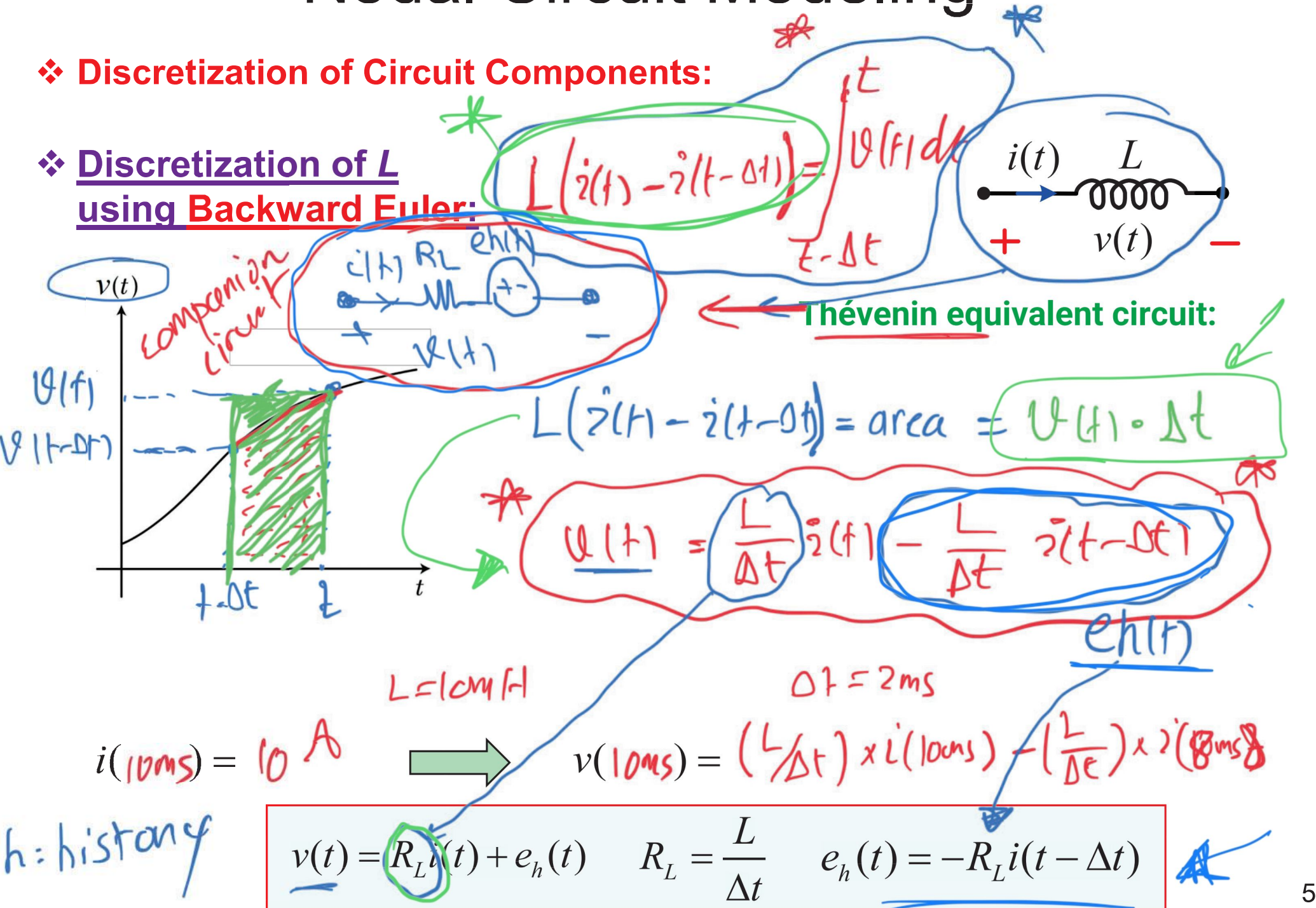


$$L i(t) \Big|_{t-\Delta t}^t = \int_{t-\Delta t}^t V(t) dt$$

Nodal Circuit Modeling

❖ Discretization of Circuit Components:

❖ Discretization of L using Backward Euler:



Nodal Circuit Modeling

❖ Discretization of Circuit Components:

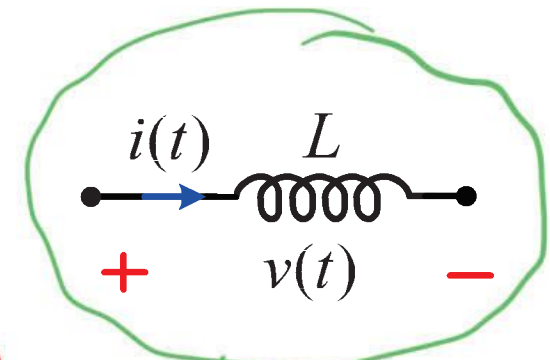
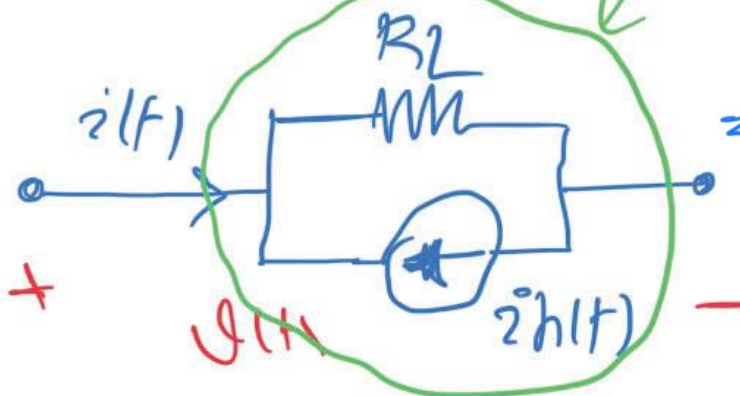
❖ Discretization of L using Backward Euler:

$$v(t) = R_L \dot{i}(t) + e_h(t)$$

$$\begin{cases} R_L = L/\Delta t \\ e_h(t) \approx -R_L \dot{i}(t - \Delta t) \end{cases}$$

$$\dot{i}_h(t) = -\dot{i}(t - \Delta t)$$

Norton equivalent circuit:

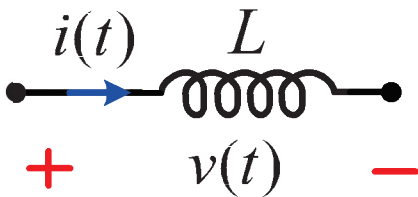
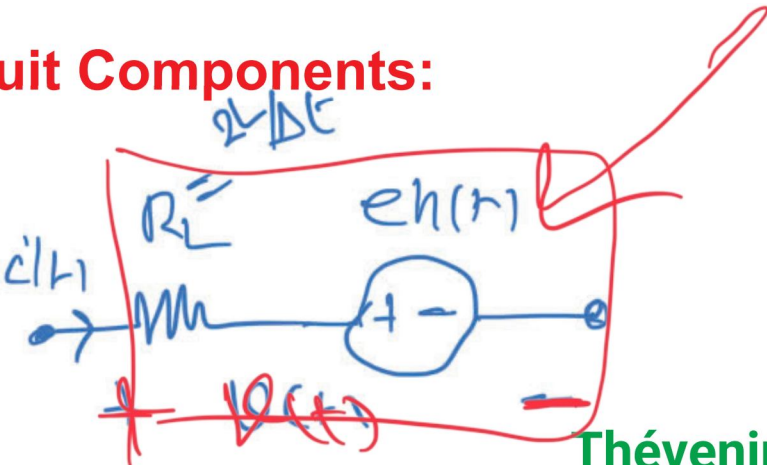


$$i(t) = \frac{1}{R_L} v(t) - i_h(t) \quad R_L = \frac{L}{\Delta t} \quad i_h(t) = -i(t - \Delta t)$$

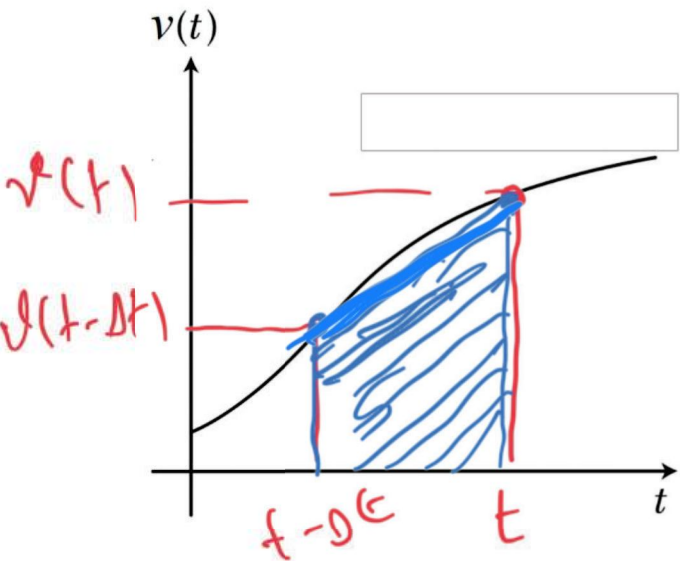
Nodal Circuit Modeling

❖ Discretization of Circuit Components:

❖ Discretization of L using Trapezoidal:



Thévenin equivalent circuit:



$$L \dot{i}(t) - L \dot{i}(t - \Delta t) = \text{area} = \frac{v(t) + v(t - \Delta t)}{2 \times \Delta t}$$

$$\left(\frac{2L}{\Delta t} \right) i(t) - \frac{2L}{\Delta t} i(t - \Delta t) - v(t - \Delta t) = v(t)$$

$$v(t) = R_L i(t) + \underbrace{\left(-R_L i(t - \Delta t) - v(t - \Delta t) \right)}_{e_h(t)}$$

$v(t) = R_L i(t) + e_h(t)$

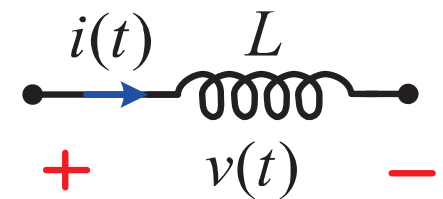
$R_L = \frac{2L}{\Delta t}$

$e_h(t) = -R_L i(t - \Delta t) - v(t - \Delta t)$

Nodal Circuit Modeling

❖ Discretization of Circuit Components:

❖ Discretization of L using Trapezoidal:



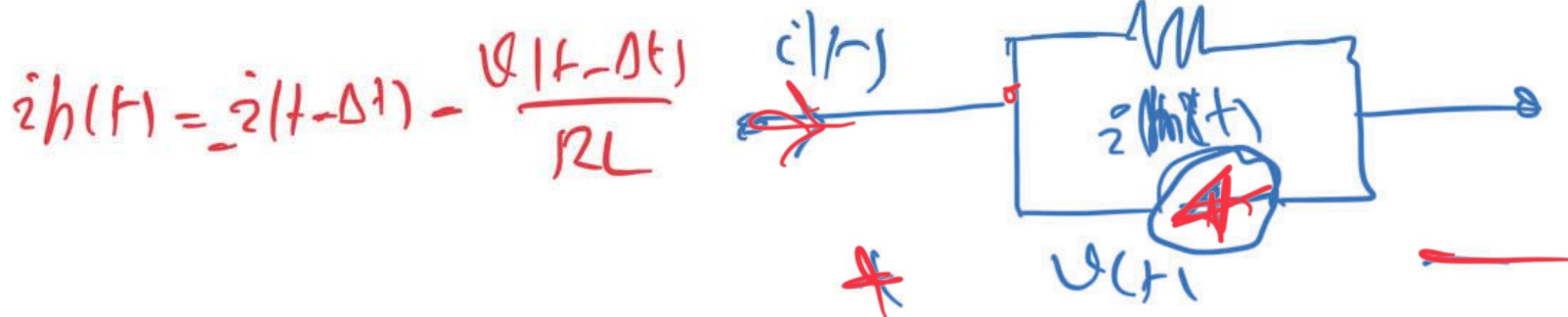
$$v(t) = R_L i(t) + e_h(t)$$

$$R_L = \frac{2L}{\Delta t}$$

$$e_h(t) = -R_L i(t - \Delta t) - v(t - \Delta t)$$

Norton equivalent circuit:

$$i_h(t) = \frac{1}{R_L} v(t) - \frac{e_h(t)}{R_L}$$



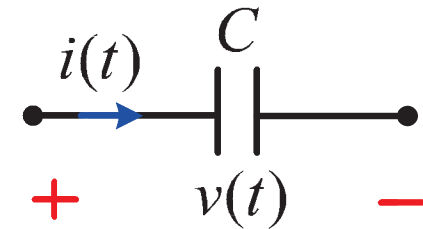
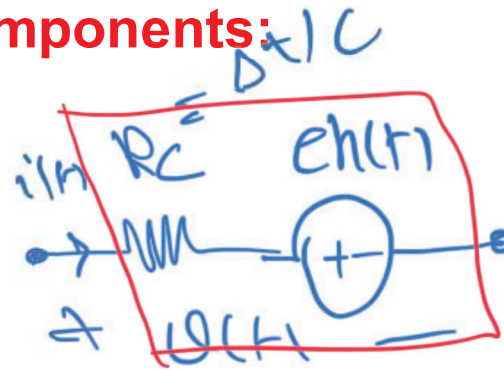
$$i_h(t) = i(t - \Delta t) - \frac{v(t - \Delta t)}{R_L}$$

$$i(t) = \frac{1}{R_L} v(t) - i_h(t) \quad R_L = \frac{2L}{\Delta t} \quad i_h(t) = -i(t - \Delta t) - \frac{v(t - \Delta t)}{R_L}$$

Nodal Circuit Modeling

❖ Discretization of Circuit Components:

❖ Discretization of C using Backward Euler:



Thévenin equivalent circuit:

$$i(t) = C \frac{dv(t)}{dt}$$

$$\int_{t-\Delta t}^t i(t) dt = \int_{t-\Delta t}^t C dv(t) \Rightarrow \text{area under } i(t) \text{ curve} = C(v(t) - v(t-\Delta t))$$

$$i(t) \Delta t = C v(t) - C v(t-\Delta t)$$

$$\Rightarrow v(t) = \frac{\Delta t}{C} i(t) + \underbrace{v(t-\Delta t)}_{e_h(t)}$$

$$v(t) = R_C i(t) + e_h(t) \quad R_C = \frac{\Delta t}{C} \quad e_h(t) = v(t - \Delta t)$$

Nodal Circuit Modeling

❖ Discretization of Circuit Components:

❖ Discretization of C using Backward Euler:

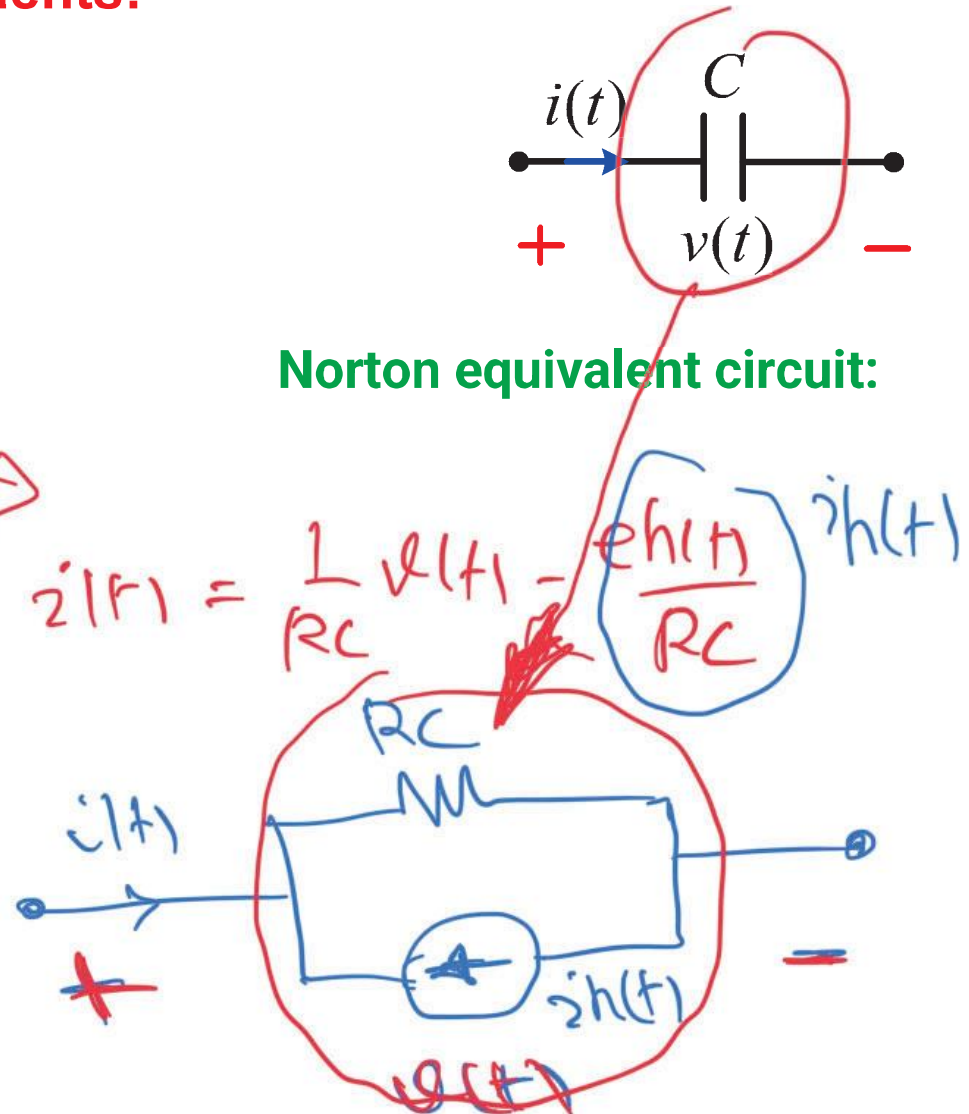
$$Q(t) = R_C \dot{Q}(t) + e_h(t)$$

$$R_C = \frac{\Delta t}{C}$$

$$e_h(t) = v(t - \Delta t)$$

$$\dot{Q}_h(t) = \frac{v(t - \Delta t)}{R_C}$$

Norton equivalent circuit:

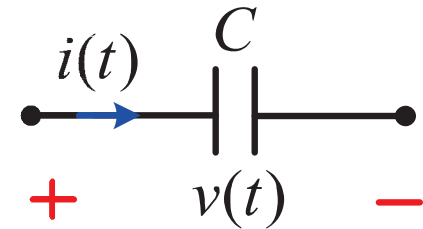
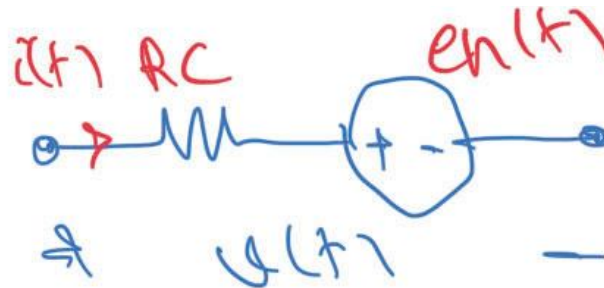


$$i(t) = \frac{1}{R_C} v(t) - i_h(t) \quad R_C = \frac{\Delta t}{C} \quad i_h(t) = \frac{v(t - \Delta t)}{R_C}$$

Nodal Circuit Modeling

❖ Discretization of Circuit Components:

❖ Discretization of C using Trapezoidal:



Thévenin equivalent circuit:

$$C \frac{dv}{dt} = i$$

$$C dv = i dt$$

$$C (v(t) - v(t - \Delta t)) = \text{area} = \frac{i(t) + i(t - \Delta t)}{2} \Delta t$$

$$v(t) = \frac{\Delta t}{2C} i(t) + \frac{\Delta t}{2C} i(t - \Delta t) + v(t - \Delta t)$$

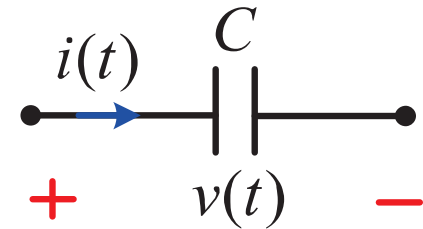
$$e_h(t)$$

$$v(t) = R_C i(t) + e_h(t) \quad R_C = \frac{\Delta t}{2C} \quad e_h(t) = R_C i(t - \Delta t) + v(t - \Delta t)$$

Nodal Circuit Modeling

❖ Discretization of Circuit Components:

❖ Discretization of C using Trapezoidal:



$$v(t) = R_C i(t) + e_h(t)$$

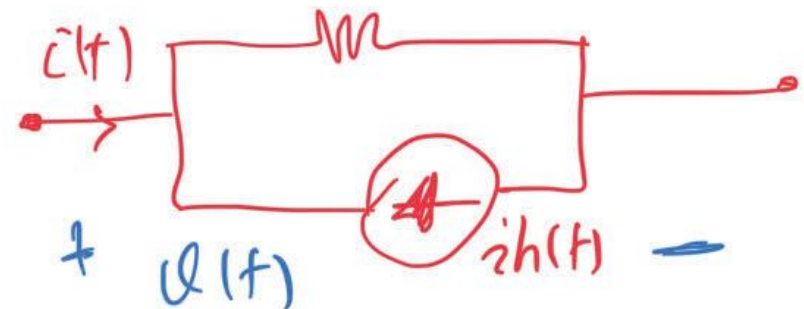
$$R_C = \Delta t / 2C$$

$$e_h(t) = R_C i(t - \Delta t) + v(t - \Delta t)$$

$$i_h(t) = \frac{e_h(t)}{R_C} = i(t - \Delta t) + \frac{v(t - \Delta t)}{R_C}$$

Norton equivalent circuit: $i_h(t)$

$$i(t) = \frac{1}{R_C} v(t) - \frac{e_h(t)}{R_C}$$

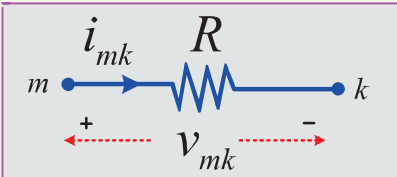
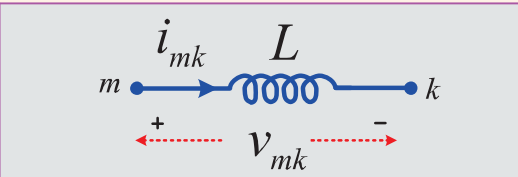
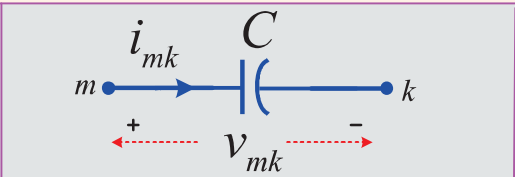
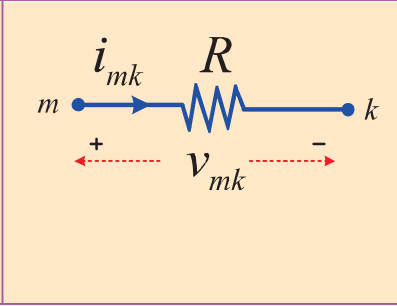
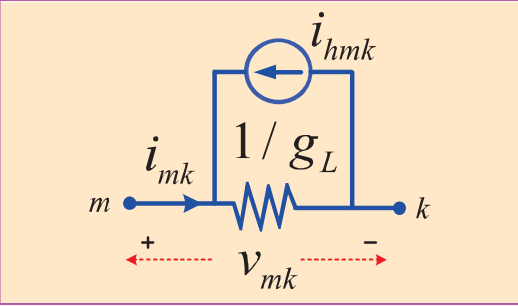
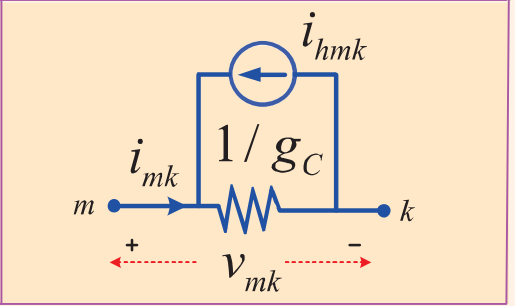


$$i(t) = \frac{1}{R_C} v(t) - i_h(t) \quad R_C = \frac{\Delta t}{2C} \quad i_h(t) = i(t - \Delta t) + \frac{v(t - \Delta t)}{R_C}$$

Nodal Circuit Modeling

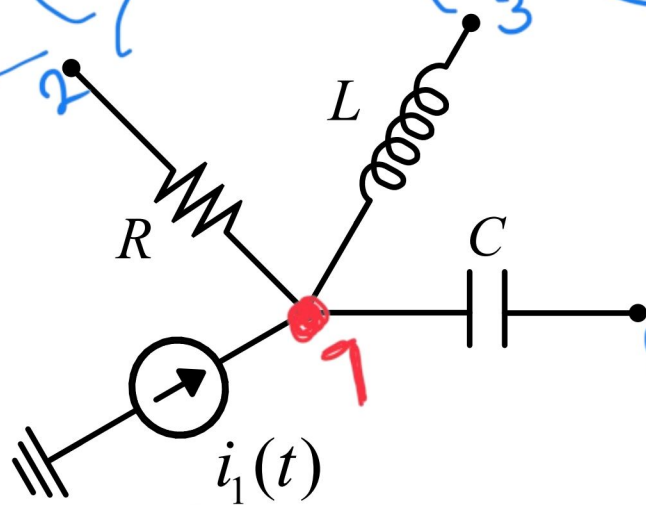
Trapezoidal

❖ Discretization of Circuit Components:

			
Continuous Time (a)	$i_{mk}(t) = \frac{1}{R} v_{mk}(t)$	$i_{mk}(t) = \frac{1}{L} \int v_{mk}(t) dt$	$i_{mk}(t) = C \frac{dv_{mk}(t)}{dt}$
Discrete Time (b)	$i_{mk}(t) = g_R v_{mk}(t)$ $g_R = \frac{1}{R}$	$i_{mk}(t) + i_{hmk} = g_L v_{mk}(t)$ $\begin{cases} g_L = \frac{\Delta t}{2L} \\ i_{hmk} = -[g_L v_{mk}(t - \Delta t) + i_{mk}(t - \Delta t)] \end{cases}$	$i_{mk}(t) + i_{hmk} = g_C v_{mk}(t)$ $\begin{cases} g_C = \frac{2C}{\Delta t} \\ i_{hmk} = [g_C v_{mk}(t - \Delta t) + i_{mk}(t - \Delta t)] \end{cases}$
Discrete Time Equivalent Circuit (c)			

Nodal Circuit Solution

❖ Network Solution:



$$\mathbf{G} \mathbf{v} = \mathbf{i}$$

$$\begin{bmatrix} \frac{1}{R} + \frac{C}{\Delta t} & -\frac{1}{R} & -\frac{C}{\Delta t} & -\frac{C}{\Delta t} \\ -\frac{1}{R} & \frac{1}{R} & 0 & 0 \\ -\frac{C}{\Delta t} & 0 & \frac{C}{\Delta t} & 0 \\ -\frac{C}{\Delta t} & 0 & 0 & \frac{C}{\Delta t} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} i_1(t) \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\mathbf{G}$ is a 4×4 matrix.

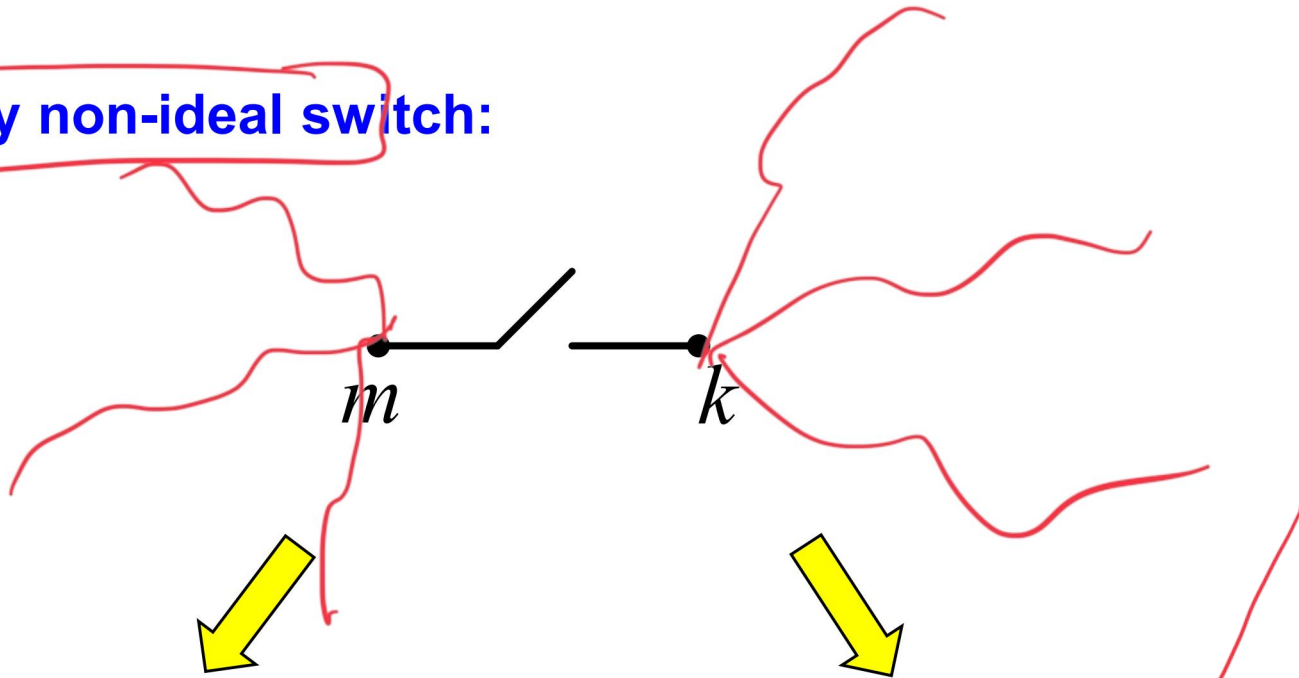
$$= \begin{bmatrix} i_1(t) + i_{hL} + i_{hC} \\ 0 \\ -i_{hL} \\ -i_{hC} \end{bmatrix} = \mathbf{i}_h$$

$$\boxed{\mathbf{G}[\mathbf{v}] = \mathbf{i}_h}$$

Nodal Circuit Solution

□ Switching Modeling:

❖ Binary non-ideal switch:



Switch ON



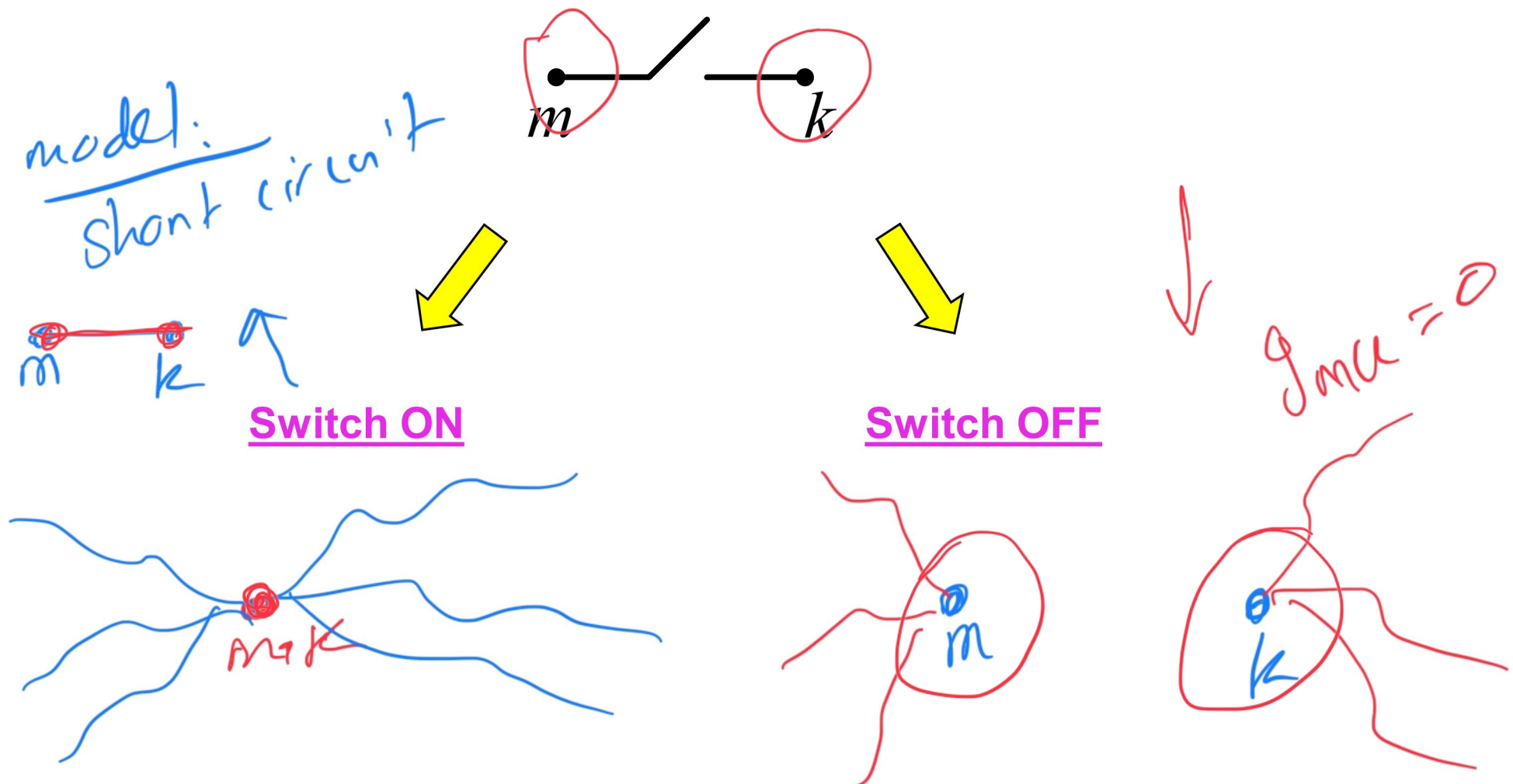
Switch OFF



Nodal Circuit Solution

❑ Switching Modeling:

❖ Ideal Switch:



Nodal Circuit Solution

❑ Switching Modeling:

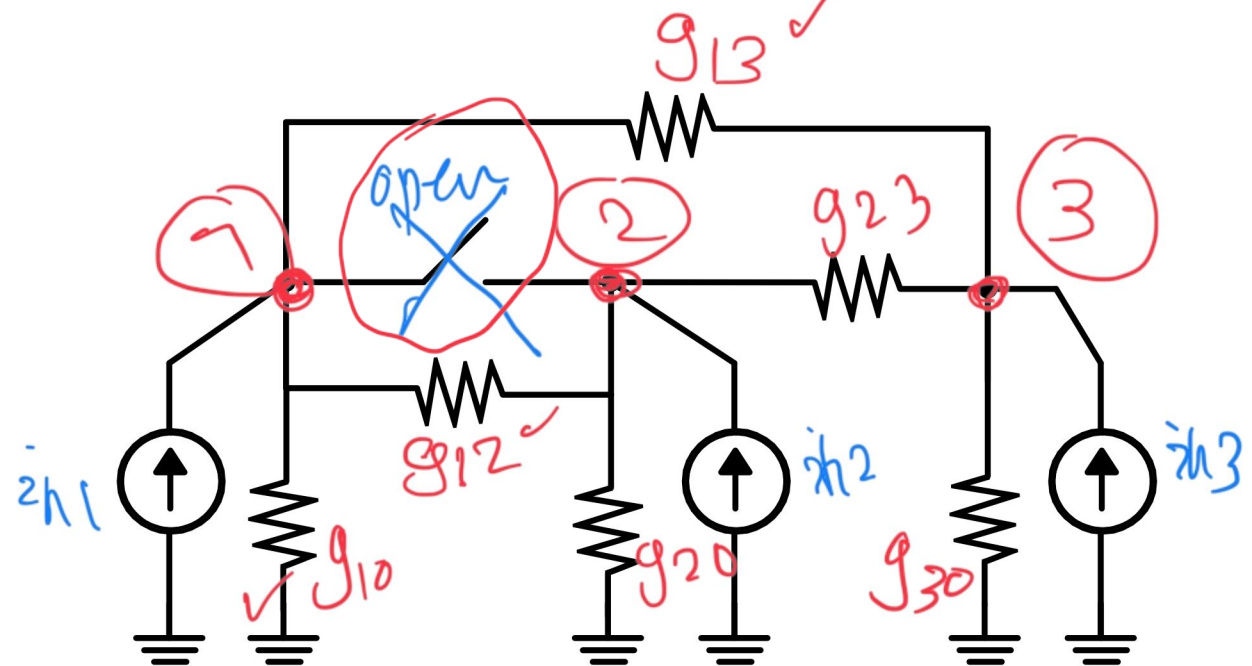
❖ Ideal Switch:

❖ Example:

$$G \cdot V = I_h$$

➤ Switch OFF:

Open



$$\begin{bmatrix}
 1 \\
 2 \\
 3
 \end{bmatrix}
 \begin{bmatrix}
 (g_{10} + g_{12} + g_{13}) & -g_{12} & -g_{13} \\
 -g_{12} & (g_{20} + g_{12} + g_{23}) & -g_{23} \\
 -g_{13} & -g_{23} & (g_{30} + g_{13} + g_{23})
 \end{bmatrix}
 \begin{bmatrix}
 V_1 \\
 V_2 \\
 V_3
 \end{bmatrix}
 =
 \begin{bmatrix}
 i_{h1} \\
 i_{h2} \\
 i_{h3}
 \end{bmatrix}$$

*

Nodal Circuit Solution

Switching Modeling:

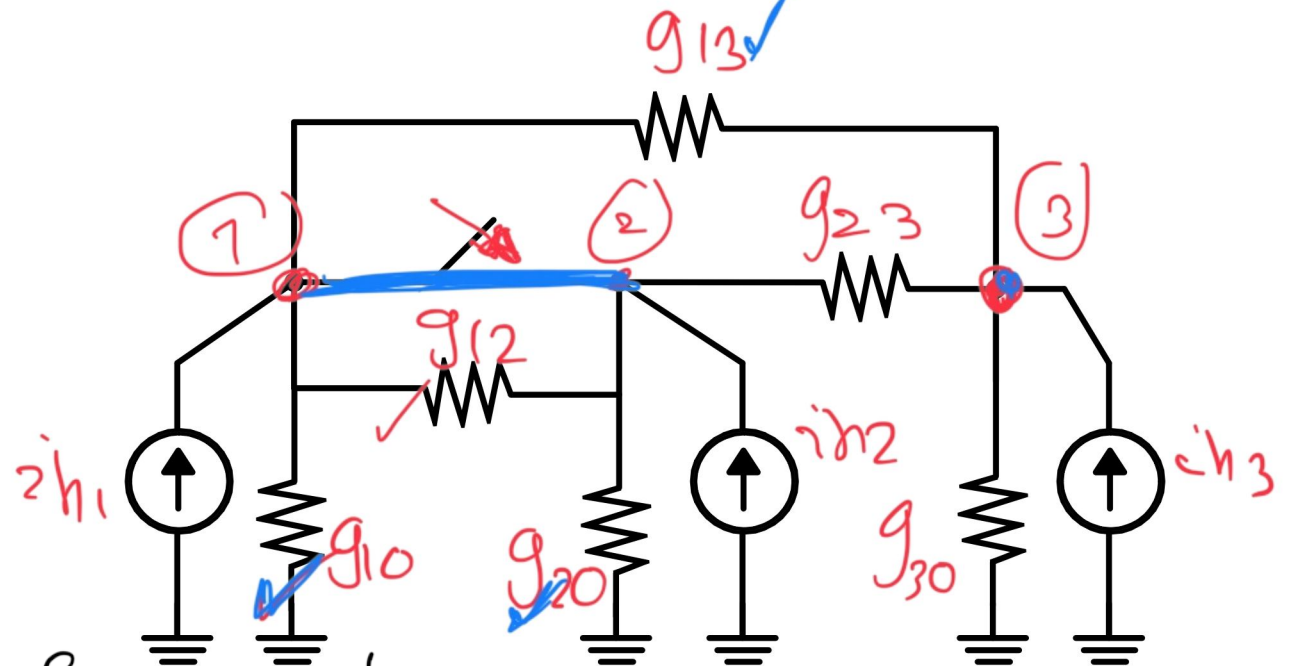
❖ Ideal Switch:

❖ Example:

$$V_2 = V_1$$

➤ Switch ON:

add row 1 & 2 in G before closing!



$$\begin{bmatrix}
 (g_{10} + \cancel{g_{12}} + g_{13} + g_{20} + g_{23}) & -(g_{13} + g_{23}) \\
 -(g_{13} + g_{23}) & (g_{30} + g_{13} + g_{23})
 \end{bmatrix}
 \begin{bmatrix}
 V_1 \\
 V_3
 \end{bmatrix}
 =
 \begin{bmatrix}
 i_{h1} + i_{h2} \\
 i_{h3}
 \end{bmatrix}$$

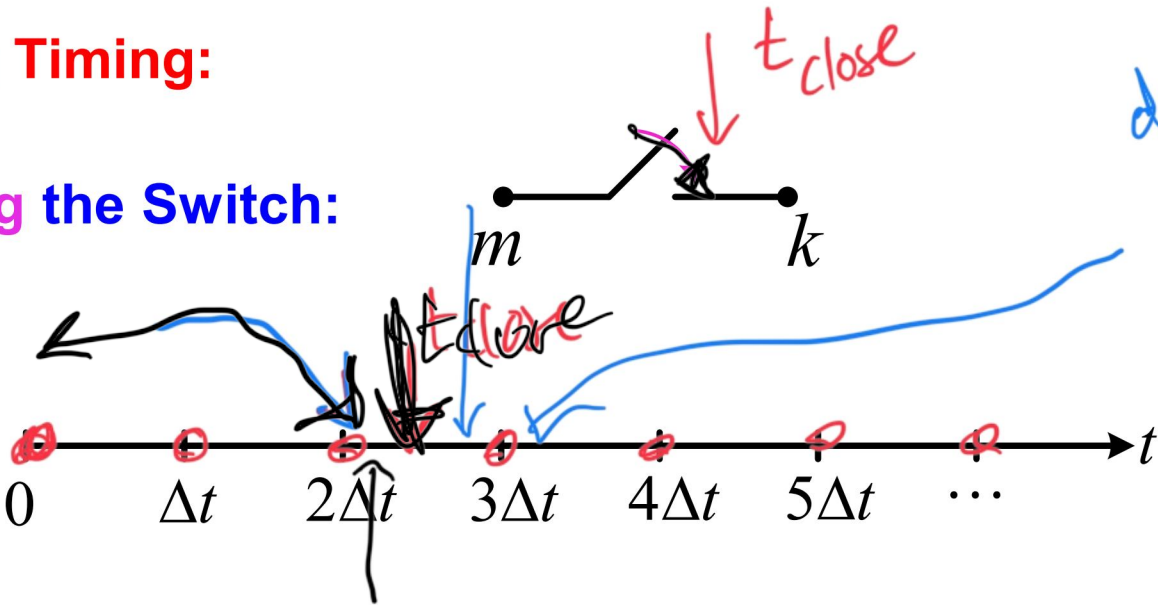
add column 1 & 2

Nodal Circuit Solution

Switching Timing:

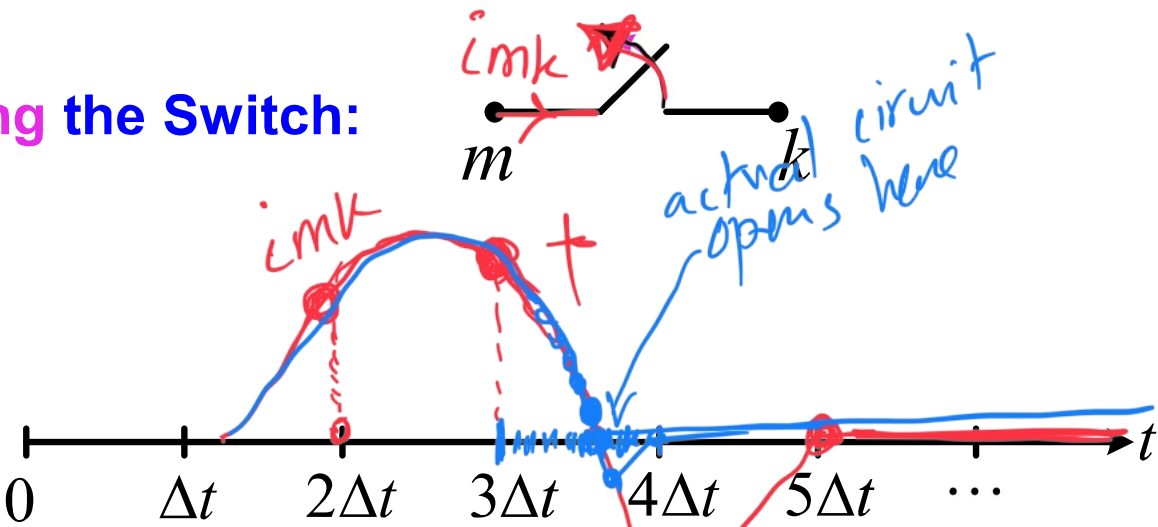
Closing the Switch:

assuming switch is open!



do calc. with removing switch is closed?
depending on how close t_{close} is to $2\Delta t$ or $3\Delta t$

Opening the Switch:



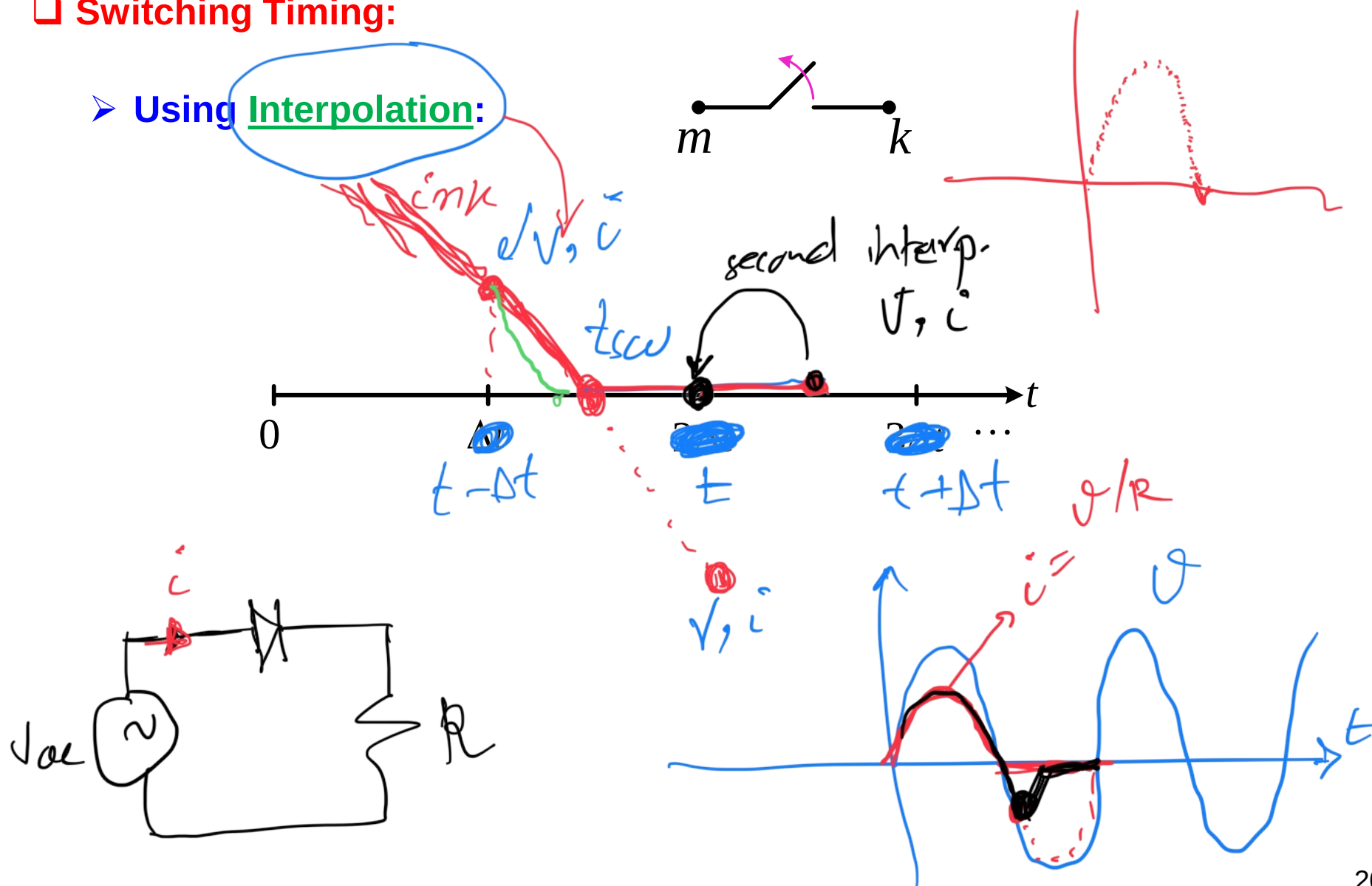
actual circuit opens here

you did your calc. assuming switch was closed!

Nodal Circuit Solution

□ Switching Timing:

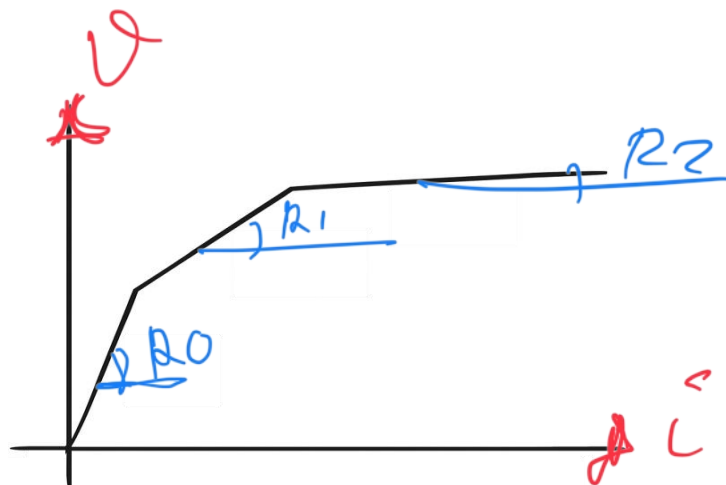
➤ Using Interpolation:



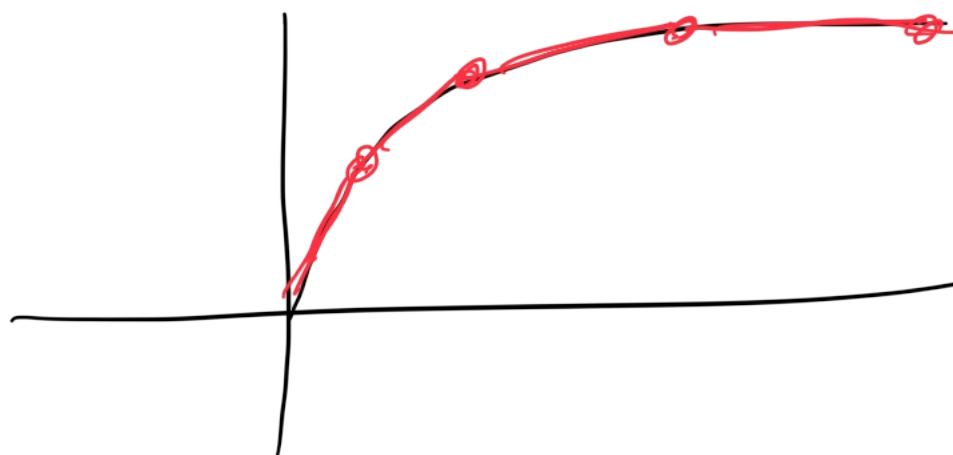
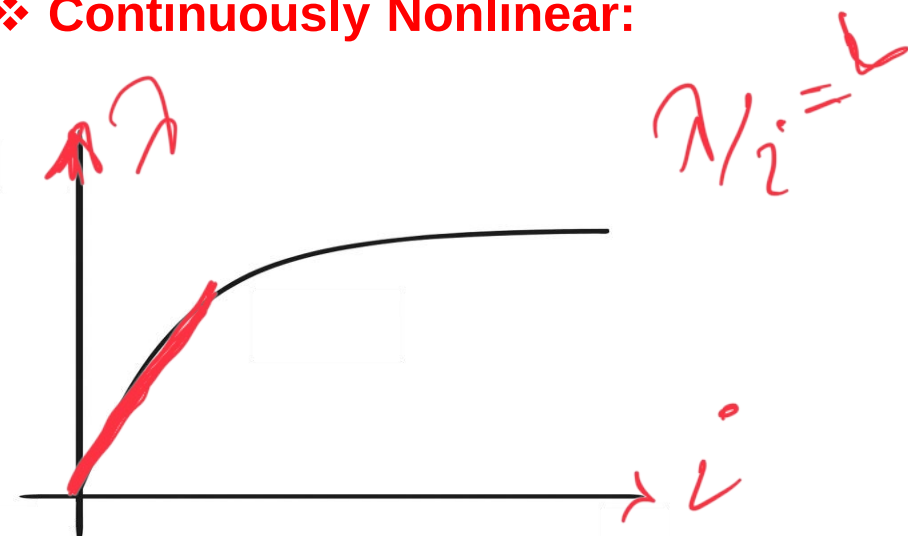
Nodal Circuit Solution

□ Modeling Nonlinear Components:

❖ Piecewise Linear:



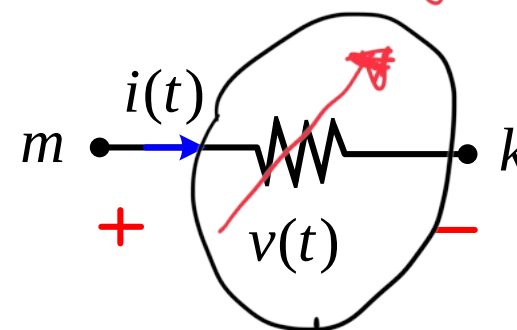
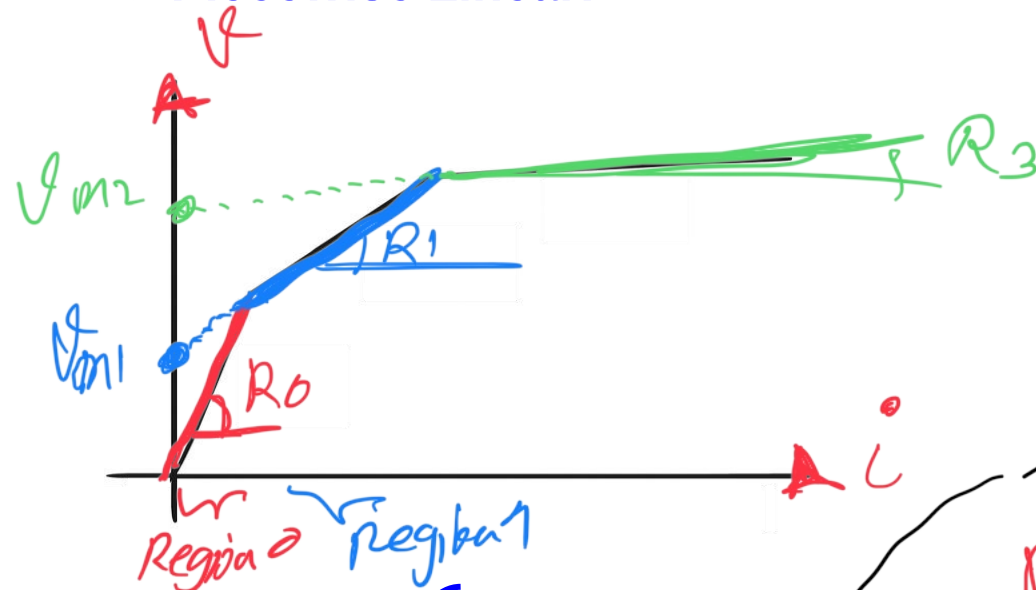
❖ Continuously Nonlinear:



Nodal Circuit Solution

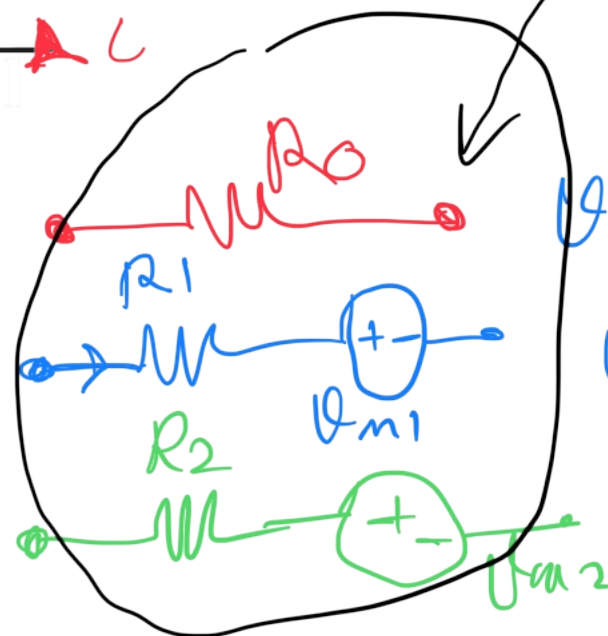
□ Modeling Nonlinear Components:

❖ Piecewise Linear:



$$v = R \cdot i$$

Region detection



$$v = R_0 i$$

$$v = R_1 i + V_{m1}$$

$$v = R_2 i + V_{m2}$$

Method 1:

Region 0:

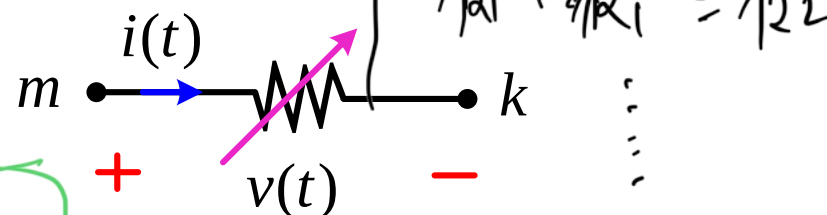
Region 1:

Region 2:

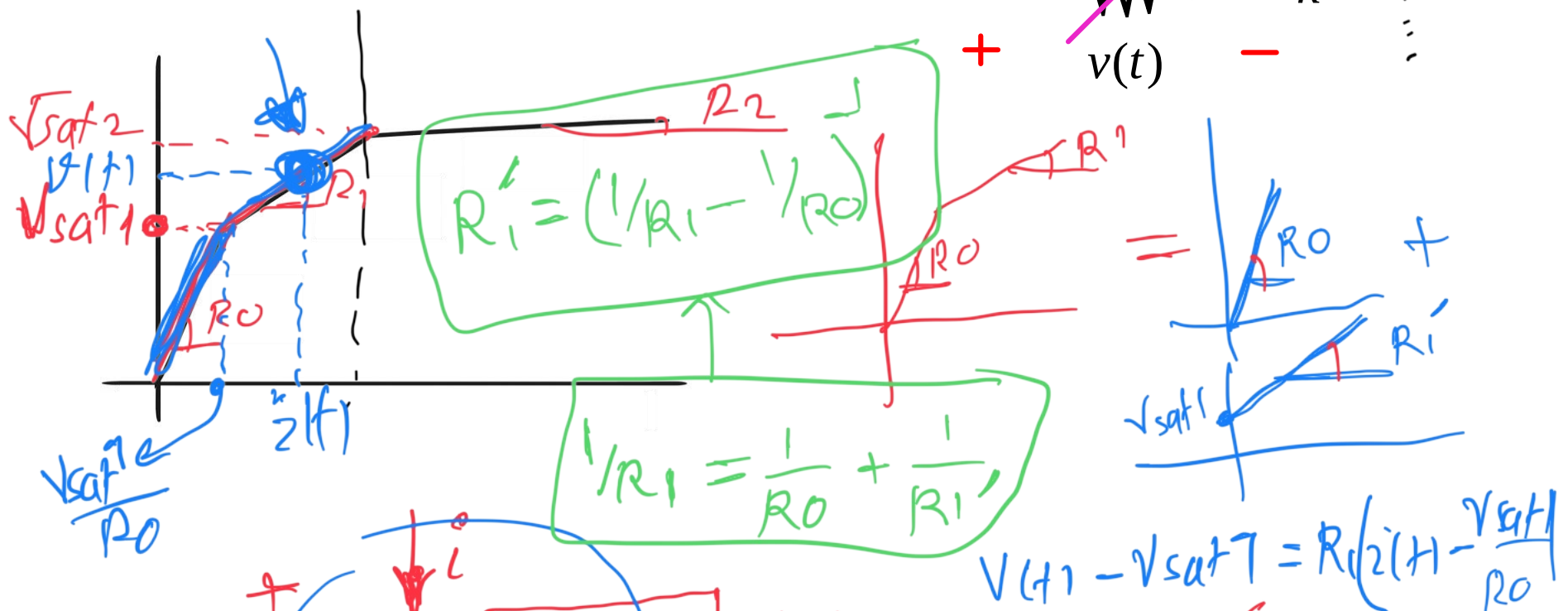
Nodal Circuit Solution

□ Modeling Nonlinear Components:

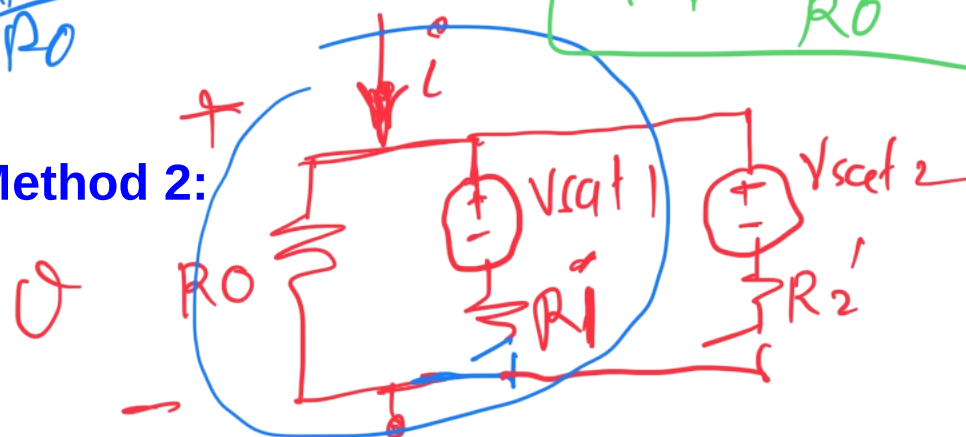
❖ Piecewise Linear:



$$\begin{cases} \frac{1}{R_0} + \frac{1}{R_1'} = \frac{1}{R_1} \\ \frac{1}{R_1} + \frac{1}{R_2'} = \frac{1}{R_2} \end{cases}$$



Method 2:

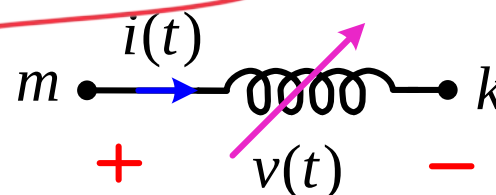
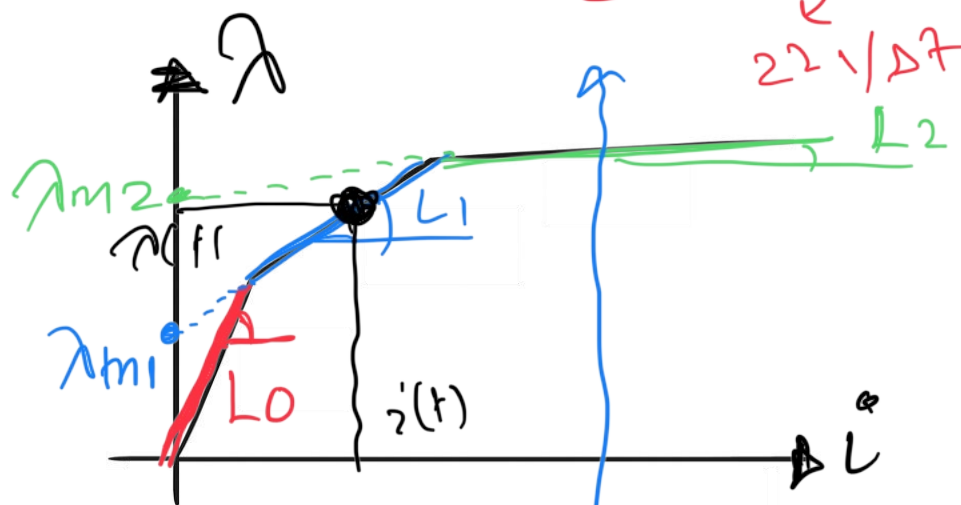


$$\begin{aligned} i(t) &= \frac{v(t)}{R_1} - \frac{v_{sat1}}{R_1} + \frac{v_{sat1}}{R_0} \\ &= \frac{v(t)}{R_0} + \frac{v(t) - v_{sat1}}{R_1'} \end{aligned}$$

Nodal Circuit Solution

□ Modeling Nonlinear Components:

❖ Piecewise Linear:



$$\lambda = L_0 i$$

$$v(t) = \frac{d\lambda(t)}{dt}$$

$$\int_{t-\Delta t}^t \lambda(t) dt = \int_{t-\Delta t}^t v(t) dt$$

Trap.

$$\lambda(t) = \lambda(t-\Delta t) + \frac{\Delta t}{2} (v(t) + v(t-\Delta t))$$

$$\lambda(t) = L_1 i(t) + \lambda_{m1} \Rightarrow L_1 i(t) + \lambda_{m1} = L_1 i(t-\Delta t) + \lambda_{m1} + \frac{\Delta t}{2} v(t) + \frac{\Delta t}{2} v(t-\Delta t)$$

$$\Rightarrow v(t) = \frac{2L_1}{\Delta t} i(t) + \frac{2}{\Delta t} \lambda_{m1} + \frac{2}{\Delta t} \lambda(t-\Delta t) + v(t-\Delta t)$$

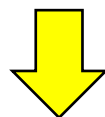
Nodal Circuit Solution

❖ Network Solution:

➤ At each simulation time-step, solve for:

$$[\mathbf{G}][\mathbf{v}] = [\mathbf{i}_h]$$

✓ The voltages of the nodes connected to voltage sources are known!



✓ Suppose nodes **B** have voltage sources

➤ Structure the matrices and vectors as:

$$\begin{bmatrix} \mathbf{G}_{AA} & \mathbf{G}_{AB} \\ \mathbf{G}_{BA} & \mathbf{G}_{BB} \end{bmatrix} \begin{bmatrix} \mathbf{v}_A \\ \mathbf{v}_B \end{bmatrix} = \begin{bmatrix} \mathbf{i}_{hA} \\ \mathbf{i}_{hB} \end{bmatrix}$$

Handwritten notes: "unknown" in red above $\mathbf{v}_A$, "known" in red below $\mathbf{v}_B$. Blue circles highlight $\mathbf{G}_{AA}$, $\mathbf{G}_{AB}$, $\mathbf{v}_A$, and $\mathbf{i}_{hA}$. A red circle highlights $\mathbf{i}_{hB}$.

$$\underline{\mathbf{G}_{AA} \cdot \mathbf{v}_A} + \mathbf{G}_{AB} \cdot \mathbf{v}_B = \mathbf{i}_{hA}$$

Handwritten notes: Blue arrows point from $\mathbf{G}_{AB}$ and $\mathbf{v}_B$ to the boxed equation below. A red arrow points from $\mathbf{v}_B$ to the boxed equation. A red arrow points from $\mathbf{i}_{hA}$ to the boxed equation.

$$A \mathbf{x} = \mathbf{b}$$

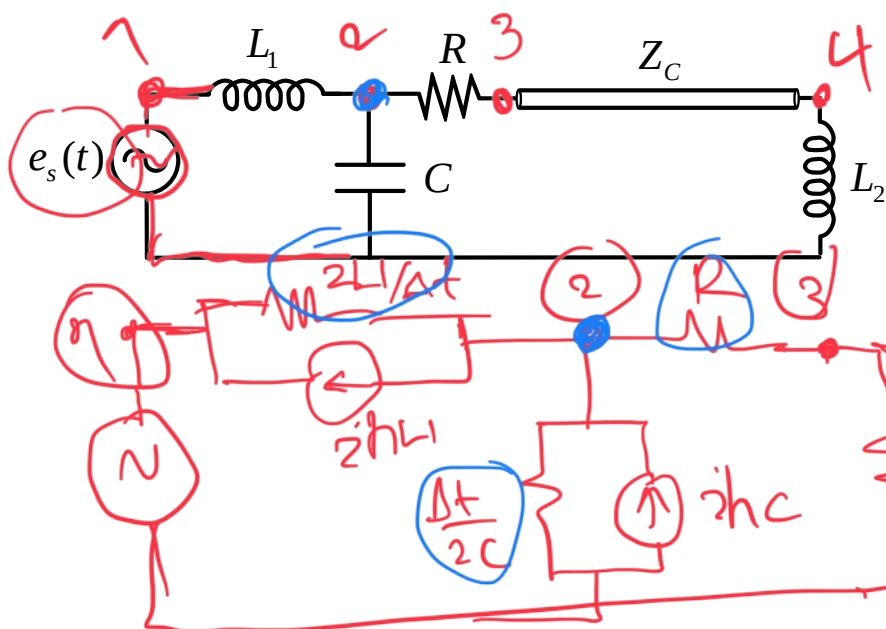
$$[\mathbf{G}_{AA}][\mathbf{v}_A] = [\mathbf{i}_{hA}] - [\mathbf{G}_{AB}][\mathbf{v}_B]$$

Handwritten notes: Red circles highlight $\mathbf{G}_{AA}$, $\mathbf{v}_A$, $\mathbf{i}_{hA}$, and $\mathbf{G}_{AB}$. A red box encloses the entire equation.

Nodal Circuit Solution

❖ Network Solution:

❖ Example:



$$\begin{bmatrix} G_{AA} & G_{AB} \\ G_{BA} & G_{BB} \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} i_{hA} \\ i_{hB} \end{bmatrix}$$

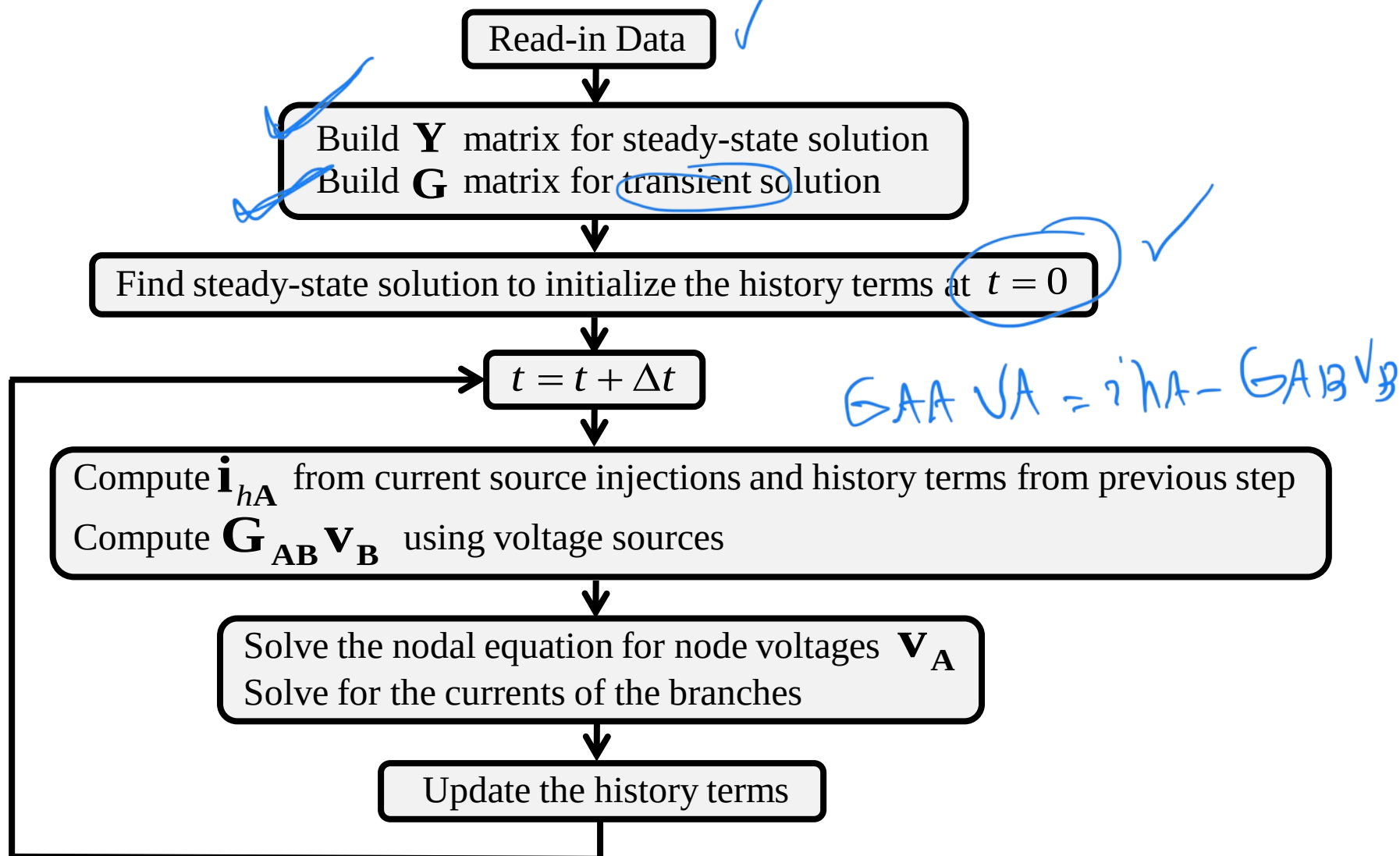
$$\begin{bmatrix} g_{22} & g_{23} & g_{24} \\ g_{32} & g_{33} & g_{34} \\ g_{42} & g_{43} & g_{44} \end{bmatrix} \begin{bmatrix} V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} i_{h2} \\ i_{h3} \\ i_{h4} \end{bmatrix}$$

$$\begin{bmatrix} \frac{\Delta t}{2L_1} + \frac{1}{Z_C} + \frac{1}{R} & 0 & 0 \\ -\frac{1}{R} & \frac{1}{R} + \frac{1}{Z_C} & 0 \\ 0 & 0 & \frac{1}{Z_C} + \frac{\Delta t}{2L_2} \end{bmatrix} \begin{bmatrix} V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} i_{hC} - i_{hL_1} \\ i_{h3} \\ i_{h4} + i_{hL_2} \end{bmatrix} - \begin{bmatrix} -\frac{\Delta t}{2L_1} \\ 0 \\ 0 \end{bmatrix} e_s(t)$$

$$\boxed{[G_{AA}][V_A] = [i_{hA}] - [G_{AB}][V_B]}$$

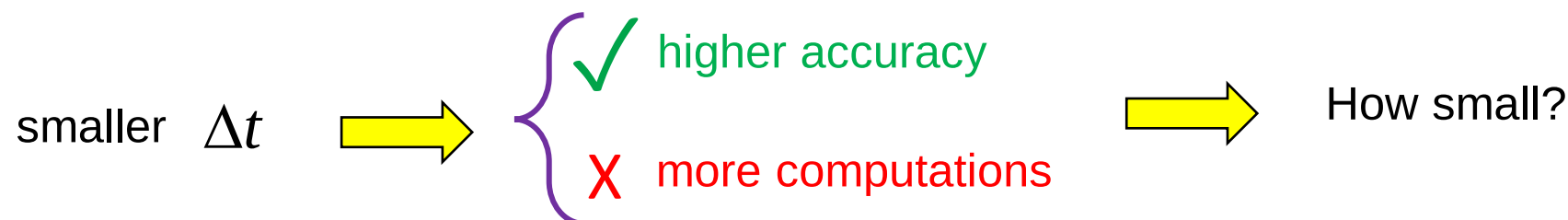
Nodal Circuit Solution

❖ Network Solution:



Nodal Circuit Solution

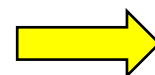
□ Time-Step Size Selection:



❖ Nyquist Sampling Rate:

- the minimum rate at which a continuous-time signal must be sampled in order to be reconstructed exactly without losing information

if a signal has a maximum
frequency component of $f_{\max}$



f_{sample}

we also need to consider the
accuracy of the integration rule!

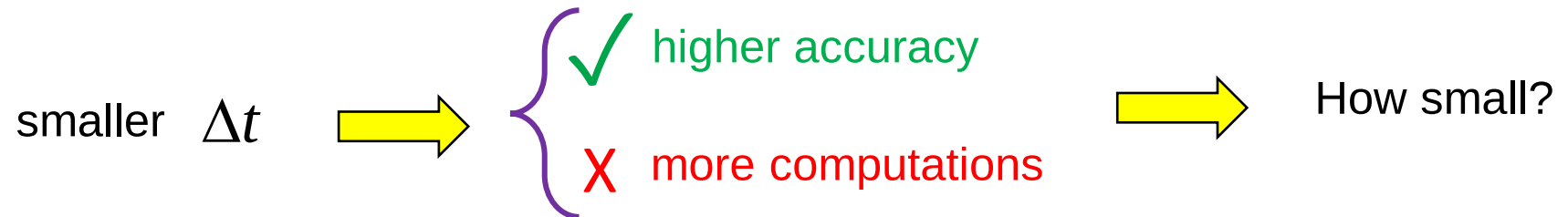
Δt



for 3% error with Trapezoidal rule $\Delta t =$

Nodal Circuit Solution

□ Time-Step Size Selection:



❖ Example:

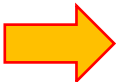
- What should be our simulation time-step size if we use the Trapezoidal integration rule and want to capture the transients up to 5 kHz with an accuracy of 97%?

Nodal Circuit Solution

❖ Network Solution:

❑ Determinant-based Cramer's Rule:

- Assume a system of n linear equations with n unknowns written in matrix form

 $\mathbf{Ax} = \mathbf{b}$

Solution 1: $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$  first calculating $\mathbf{A}^{-1}$ and then multiplying by $\mathbf{b}$

Solution 2: $\mathbf{x} = \mathbf{A} \setminus \mathbf{b}$  directly calculating $\mathbf{x}$ using left division (\)

❑ In EMTP-type Programs:

- They use Gaussian elimination and LU decomposition method instead of determinant-based Cramer's rules

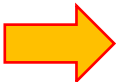
Number of Nodes	Gaussian Elimination	Cramer's Rule Straightforward	Cramer's Rule with Laplace Expansion
5	1.5 s	1 min	4 s
10	10 s	85 days	4 min
20	1 min	10^{11} years	4.6 days

Nodal Circuit Solution

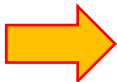
❖ Network Solution:

□ Determinant-based Cramer's Rule:

- Assume a system of n linear equations with n unknowns written in matrix form


$$\mathbf{Ax} = \mathbf{b}$$

- ✓ For each unknown x_i replace the i^{th} column of $\mathbf{A}$ with $\mathbf{b}$. Call this new matrix $\mathbf{A}_i$. Compute the unknown as:

 We need to calculate the determinant of _____ matrices _____ times!

$$x_i = \frac{\det(\mathbf{A}_i)}{\det(\mathbf{A})}$$



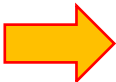
order of computations:

Nodal Circuit Solution

❖ Network Solution:

❑ Gaussian Elimination:

- Assume a system of n linear equations with n unknowns written in matrix form


$$\mathbf{Ax} = \mathbf{b}$$

- Forward Elimination:

- Use row operations to make the entries under the diagonal zero

- Backward Substitution:

- Solve for the last unknown x_n and substitute in the upper rows to calculate $x_{n-1}, \dots, x_1$



order of computations:

Nodal Circuit Solution

❖ Network Solution:

❑ Gaussian Elimination:

❖ Example:

➤ Solve:

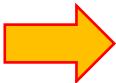
$$\begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -11 \\ -3 \end{bmatrix}$$

Nodal Circuit Solution

❖ Network Solution:

❑ LU Decomposition:

- Assume a system of n linear equations with n unknowns written in matrix form


$$\mathbf{Ax} = \mathbf{b}$$

- ✓ Assume that we can express the matrix $\mathbf{A}$ as the $\mathbf{A} = \mathbf{LU}$ where $\mathbf{L}$ is an a Lower-triangular matrix and $\mathbf{U}$ is and Upper-triangular matrix

Nodal Circuit Solution

❖ Transmission Line Modeling:

To be continued...